

Video Editing Chatbot: Language-Driven Video Compositing System

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Abstract

In this work, we present a video editing chatbot (VEC) that performs intelligent multimedia editing through natural language dialogue. VEC comprises three modules: instruction analysis, multimedia resources retrieval, and multimedia resources editing. It analyzes user instructions to retrieve relevant multimedia resources from the multimedia database (MMDB), and then applies appropriate editing methods from the multimedia toolbase (MMTB) automatically. To enhance user experience and simplify operation, VEC uses a multi-turn dialogue mechanism to handle complex editing tasks.

CCS Concepts

• **Computing methodologies** → **Natural language processing**: Visual content-based indexing and retrieval; • **Information systems** → **Data management systems**.

Keywords

Video-Text Retrieval, Vision-Language Understanding, Cross-Modal Representation

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1 Introduction

Multimedia content editing involves integrating diverse media elements like image, audio, and video using specialized software,

but it typically requires significant time and expertise. With advancements in large language models (LLMs), conversational artificial intelligence has made notable progress in text-based dialogue systems[2, 4, 13, 14]. We build a chatbot system for video editing that simplifies the process by enabling video composition through language, making it faster and easier.

The video editing chatbot(VEC) system pipeline, as shown in Figure 1, comprises three modules: instruction analysis, multimedia resources retrieval, and multimedia resources editing. The instruction analysis module uses LLMs to analyse user instructions into retrieval and editing instructions, and extract key elements. The multimedia resources retrieval module accesses a large-scale, knowledge-organized multimedia database (MMDB). Compared to traditional editing tools[1, 3], VEC supports both user-uploaded resources and its own vast collection, enhancing creative flexibility. The multimedia resources editing module applies AI media editing algorithms from the multimedia toolbase (MMTB) to perform tasks like video matting and background replacement. Compared with existing automatic editing tools[11, 12], VEC achieves fully language-driven multimedia content editing with advanced methods and scalability for various creative tasks.

2 SYSTEM ARCHITECTURE

The system architecture consists of three main modules: instruction analysis, multimedia resources retrieval, and multimedia resources editing.

2.1 Instruction Analysis

The instruction analysis module uses LLMs like ChatGPT[2] and ChatGLM[14] to translate user instructions into executable operations. It consists of three sub-modules: instruction classification, 5W1H analysis, and editing method analysis.

Instruction Classification: To improve system efficiency, VEC first divides user instructions into retrieval instructions (resources obtaining) and editing instructions (resources editing). Upon receiving instructions, LLMs analyze and classify them accordingly.

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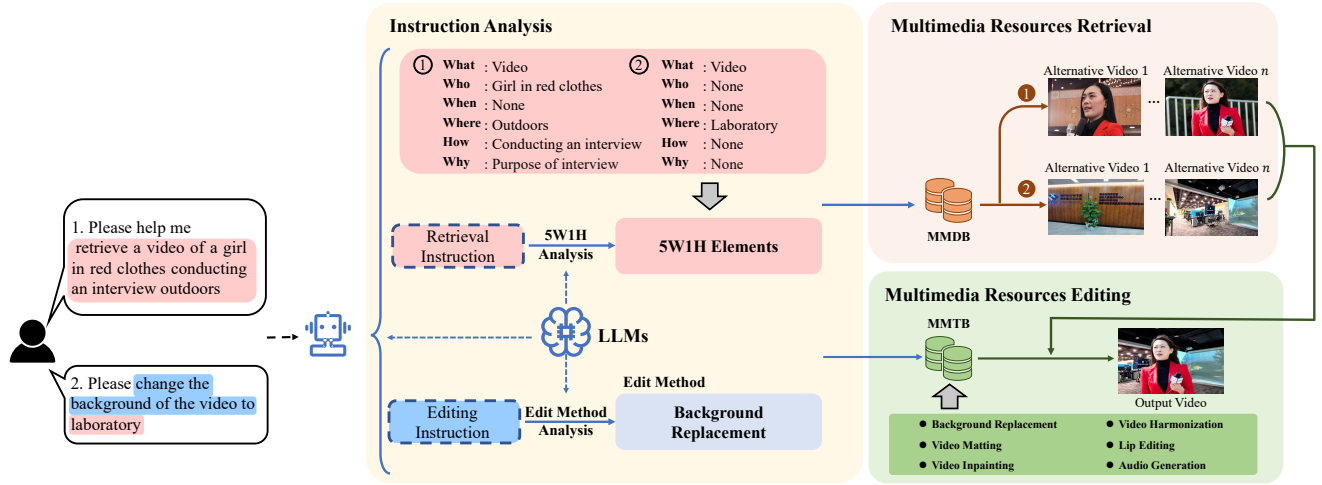


Figure 1: The pipeline of video editing chatbot system

5W1H Analysis: Retrieval instructions analysis focuses on selecting specific resources using LLMs for 5W1H analysis. The "What" element specifies the resource format, defaulting to video. This process ensures comprehensive extraction of crucial information, improving the relevance and accuracy of queries to match requirements closely. **Editing Method analysis:** For editing instructions, the focus is on specific resource edits. LLMs analyze user instructions to determine the appropriate editing methods and required parameters. If instructions lack essential details, default values are used to ensure smooth task execution.

2.2 Multimedia Resources Retrieval

Multimedia resources retrieval module comprises retrieval methods and MMDB. The retrieval method extracts 5W1H to precisely analyze user requirements and retrieves the most relevant resources from the extensive MMDB.

Retrieval Method: For audio resources, the system employs a keyword matching strategy, leveraging 5W1H analysis to match metadata like titles, descriptions, and tags. For visual resources, VEC employs ALPRO[8] as the retrieval method. ALPRO generates structured textual templates based on 5W1H, enriching semantic information and aligning textual and visual features at different levels. This ensures comprehensive matching at both coarse and fine granularity, allowing accurate selection of visual resources even in complex scenes.

Multimedia Database: MMDB integrates audio, video, and image data collected through web scraping and public databases. It utilizes extensive web-crawling resources from various platforms and filters and integrates high-quality data from public databases. MMDB supports dynamic updates to maintain the timeliness and diversity of its content.

2.3 Multimedia Resources Editing

Multimedia resources editing module automates editing operations based on user requirements using the MMTB, which integrates

multi-functional algorithms. Current MMTB addresses key user needs with six multimedia processing operations: video matting, inpainting, harmonization, background replacement, lip syncing, and audio synthesis.

VEC employs advanced, stable algorithms for efficient and reliable editing. Video matting includes portrait matting with the RVM[10] for superior transparency estimation and object matting with the Lseg[7] for precise segmentation in complex backgrounds. The Harmonizer[5] ensures video harmonization blends naturally with the target scene. Background replacement integrates matting and harmonization with user-specified backgrounds. The E2FGVI[9] achieves stable video inpainting. Lip syncing and audio synthesis are handled by the text-driven VITS[6].

MMTB's design emphasizes flexibility and scalability, allowing VEC to integrate advanced multimedia processing algorithms through preset invocation interfaces.

3 USER INTERFACE

The user interface design of video editing chatbot is shown in the demo video. It is inspired by the interactive modes of conversational AI systems such as ChatGPT. It aims for a "zero-threshold" editing experience, allowing easy multimedia editing without technical expertise.

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